

Memory-Centric Agentic Inference — cycles 7-9 clone 0

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Abstract

Cycles 7-9 closed fanout clone 0 for DC-005, the deferred constant for production trajectory reuse measurement. Earlier clone work had already produced the required artifact, `memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md`, and integrated its TRJ-001 through TRJ-007 experiment rows into the parent measurement harness. These cycles did not redesign the measurement plan. They converted the branch from “validated design” into “conductor-merge-ready package.”

The main event was a cycle 7 verifier correction. The DC-005 merge verifier originally treated any future DC-006 row in shared measurement CSVs as contamination. That was too broad because the parent harness now contains sibling constants such as DC-003 and DC-004, and it may later contain legitimate DC-006 rows. The cycle 7 audit narrowed the rule: sibling constants are allowed in shared CSVs, but DC-006 or any sibling constant must not reuse DC-005 TRJ-* experiment IDs or DC-005 collapse-threshold IDs. Source: auditor session 2e641351-22d1-4452-ae62-b4b066fc8f54.

Cycle 8 reconfirmed that the narrowed verifier preserved the DC-005 boundary. The plan artifact, required TRJ-* rows, thresholds, fields, and invalid-replay zero-positive-value rule remained present. The verifier passed 10/10 checks, `org_check` passed, and temporary probes showed that legitimate non-TRJ sibling rows passed while contaminating DC-006, TRJ-* rows failed. `promise_check` still reported fanout ledger warnings, which the auditor classified as a registration issue, not a DC-005 semantic defect. Source: auditor session e4fecfee-3f1a-4a2a-ab79-87d21226830f.

Cycle 9 completed the closure: the same semantic checks passed, `org_check` passed, and `promise_check` became green with 66 ledger events. The final decision is `VALIDATED`. Clone 0 has no remaining local research or build work. Future work belongs to conductor merge handling, and DC-005 should reopen only if a verifier run detects missing artifacts, missing TRJ-* rows, missing fields or thresholds, corrupt CSVs, invalid replay receiving positive value,

or sibling reuse of DC-005 IDs or collapse thresholds. Source: auditor session b166460d-2d8a-4a57-b87f-d79dc49e5f8c.

Introduction

The root project investigates whether future AI infrastructure should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Within that project, clone 0 was assigned one narrow branch: design the production measurement plan for trajectory reuse, recorded as deferred constant DC-005.

DC-005 covers trajectory-level state in agentic inference. A trajectory is the execution history of an agentic run: branch decisions, verifier results, tool outputs, replay edges, durable workspace dependencies, and merge/discard outcomes. A directed acyclic graph, or DAG, represents those dependencies without cycles. The DC-005 question is whether that trajectory-level state has enough retained value to justify Option C, the trajectory/DAG-aware memory fabric.

The architecture options used by the broader project are:

- **Option A:** conventional request/model/KV-centric serving.
- **Option B:** memory-object-aware runtime for objects such as retrieved context, semantic-cache entries, tool outputs, and provenance state.
- **Option C:** trajectory/DAG-aware memory fabric for branch state, verifier state, trajectory logs, replay edges, and durable workspace dependencies.

Clone 0 had already produced the core measurement design before cycles 7-9. The artifact `memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md` defines seven trajectory experiments, required production fields, threshold equations, synthetic mechanism probes, and acceptance criteria. Cycles 7-9 therefore focused on preservation: ensuring that the validated DC-005 package could survive conductor merge and coexist with sibling measurement branches.

Methodology

The cycles followed a closure pattern rather than a build pattern. Researchers stated that DC-005 was complete and should not be expanded locally. Workers confirmed that no new clone-local artifacts were needed. Auditors ran or reported the decisive checks and recorded whether the conductor could accept the branch.

The central executable artifact was `tests/verify_dc005_merge_ready.py`. The verifier reads the parent measurement harness CSVs:

- `data/measurement_experiment_specs.csv`
- `data/measurement_required_fields.csv`
- `data/measurement_thresholds.csv`
- `data/measurement_claim_update_matrix.csv`
- `data/measurement_synthetic_probe_results.csv`

It writes `data/dc005_merge_verification_results.csv`. The result file records one row per merge-readiness check.

The closure checks were:

- The DC-005 plan artifact exists.
- TRJ-001 through TRJ-007 are present.

- Required thresholds are present: `C_to_B_trajectory_reuse`, `C_to_A_trajectory_reuse`, `p_survive_min`, and `p_verifier_reuse_min`.
- Required fields are present: `trajectory_node_id`, `replay_authorization_scope`, `verifier_evidence_hash`, and `retention_hold_state`.
- Unauthorized, stale, tampered, or retention-invalid replay cannot add positive retained value.
- Sibling constants are allowed in shared measurement CSVs, but sibling rows must not reuse DC-005 trajectory experiment IDs or collapse thresholds.

Results

Cycle 7: Verifier Boundary Correction

Cycle 7 started from a terminal handoff position. The researcher session `efcfcef0-fe2c-456f-a32a-5b2cba64dbd7` stated that DC-005 was already validated and that the next correct action was conductor merge handling. The worker session `5ea411ea-62ae-4821-bfc2-7c38d063fc51` built no new clone-local artifacts and listed the validated package.

The audit then found one moderate issue in the merge verifier. The old verifier rejected any row with `deferred_constant=DC-006` in shared parent measurement CSVs. That was too broad because the parent measurement harness is multi-constant: it already contains DC-003 durable replay-tail rows and DC-004 semantic-cache risk rows alongside DC-005 rows. A future legitimate DC-006 row should not fail DC-005 verification merely because it exists in a shared CSV.

The auditor patched the verifier and its documentation. After the patch, the contamination rule became narrower and more precise:

- Legitimate sibling constants are allowed.
- A sibling row fails DC-005 verification only if it reuses a DC-005 TRJ-* experiment ID or one of the DC-005 collapse thresholds.
- Authorization and provenance fields remain validity gates for DC-005 replay, but provenance-overhead magnitude remains DC-006 scope.

The cycle 7 audit validated the patch with these checks:

Check	Result
<code>python3 tests/verify_dc005_merge_ready.py</code>	PASS, 10/10 checks
<code>python3 -m py_compile tests/verify_dc005_merge_ready.py</code>	PASS
<code>python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace</code>	PASS
<code>python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace</code>	warnings only
Temporary legitimate non-TRJ DC-006 sibling row	verifier PASS
Temporary contaminating DC-006, TRJ-* row	verifier FAIL

The decision was VALIDATED. The only remaining issue was a minor ledger-registration warning from fanout shadow-ledger isolation.

Cycle 8: Reconfirmation of the Narrowed Rule

Cycle 8 repeated the closure position with the corrected verifier as the baseline. The researcher session 6c78a9bd-a285-4953-8efe-fae3482ecd3a held DC-005 semantics constant and focused on merge-handoff contamination criteria. The worker session d3c7f726-bb64-4d63-8094-affd9877ff3a built nothing new and restated the conductor acceptance commands.

The auditor session e4fecfee-3f1a-4a2a-ab79-87d21226830f checked that the plan remained intact and that the narrowed verifier still behaved correctly. The audit found no critical or moderate findings and one minor issue: `promise_check` still reported orphan artifacts, including the verifier and verification-results file. Because the semantic verifier and `org_check` passed, the auditor classified this as a fanout or ledger-registration issue rather than a DC-005 semantic failure.

Cycle 8 validation results:

Check	Result
Plan exists and remains 298 lines	PASS
Static scan for TRJ-001 through TRJ-007	PASS
Static scan for required thresholds	PASS
Static scan for required fields	PASS
Invalid-replay zero-positive-value rule	PASS
python3 tests/verify_dc005_merge_ready.py	PASS, 10/10 checks
python3 -m py_compile tests/verify_dc005_merge_ready.py	PASS
org_check	PASS
promise_check	warnings only
Legitimate non-TRJ DC-006 sibling threshold probe	PASS
Contaminating DC-006, TRJ-999-contamination probe	FAIL as intended

The decision was again VALIDATED.

Cycle 9: Final Closure With Green Ledger State

Cycle 9 confirmed the same semantic state after conductor/root reconciliation. The researcher session 8b48e188-f3bc-45d9-9c64-3649a8de253a declared DC-005 complete and identified the remaining question as terminal merge-readiness. The worker session 82137f5f-2e2d-4c62-86a0-98d18eee9d31 built no new artifacts and noted that clone 0 had no remaining build work.

The auditor session b166460d-2d8a-4a57-b87f-d79dc49e5f8c reported the final decisive change: `promise_check` was now green with 66 ledger events. This resolved the previous minor ledger caveat. The semantic verifier still passed 10/10 checks, `org_check` passed, and the same adjacent contamination probes behaved correctly.

Cycle 9 validation results:

Check	Result
Plan contains TRJ-001 through TRJ-007	PASS
Plan contains required threshold names	PASS

Check	Result
Plan contains required security/retention fields	PASS
Plan contains invalid-replay zero-positive-value rule	PASS
python3 tests/verify_dc005_merge_ready.py	PASS, 10/10 checks
org_check	PASS
promise_check	PASS, green with 66 ledger events
Legitimate non-TRJ DC-006 sibling threshold probe	PASS
Contaminating DC-006, TRJ-999-contamination row	FAIL on no_dc006_measurement_rows

The final decision was VALIDATED.

Findings

The first finding is that DC-005 is closed at clone scope. The required measurement plan exists, the parent harness rows exist, and the verifier protects the required IDs, fields, thresholds, and scope boundary.

The second finding is that the verifier now matches the multi-constant structure of the parent harness. Shared CSVs may contain sibling constants, including future DC-006 rows. DC-005 verification fails only when a sibling row reuses DC-005-specific TRJ-* experiment IDs or DC-005 collapse thresholds.

The third finding is that authorization, provenance, verifier evidence, and retention fields are valid DC-005 requirements only as replay-validity gates. They prevent unauthorized, stale, tampered, or retention-invalid replay from receiving positive retained value. They do not move provenance-validation overhead magnitude into DC-005; that remains sibling DC-006 scope.

The fourth finding is that the earlier ledger caveat has been resolved. Cycle 7 and cycle 8 still saw promise_check warnings, but cycle 9 reported promise_check green with 66 ledger events. The conductor no longer needs to treat DC-005 verifier artifacts as unresolved orphan artifacts in the current root state.

Discussion

The purpose of cycles 7-9 was preservation rather than discovery. That matters because DC-005 is a measurement-design branch, not a production measurement result. The artifact defines how production systems should measure trajectory reuse; it does not claim that production trajectory reuse has already been measured.

The corrected verifier makes the branch more robust. Without the cycle 7 patch, future sibling work on DC-006 could have caused false DC-005 failures simply by adding legitimate rows to shared CSVs. With the patch, DC-005 protects its own identity while allowing the parent harness to grow.

The closure criteria are intentionally narrow. DC-005 should not reopen because a sibling branch adds rows to shared measurement tables. It should reopen only when the trajectory-reuse package itself is damaged or contaminated: missing plan, missing TRJ-* rows, missing

required fields, missing collapse thresholds, corrupt CSVs, invalid replay receiving positive value, or sibling reuse of DC-005 IDs or thresholds.

No figures were produced in cycles 7-9 for this clone. Existing workspace figures belong to broader root-cycle work or earlier architecture/modeling milestones, not to this terminal merge-validation pass.

Conclusions and Recommendations

Clone 0 should be accepted for conductor merge purposes. DC-005 remains complete and validated for its scoped objective: production trajectory reuse measurement covering trajectory DAG events, branch survival, verifier-state reuse, tool-output replay, trajectory-log replay, durable workspace reuse, invalid-replay gates, and Option C collapse thresholds.

The conductor acceptance command set is:

```
python3 tests/verify_dc005_merge_ready.py
python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace
python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace
```

Accept DC-005 when the verifier passes, org_check passes, and promise_check remains green or reports only non-semantic conductor ledger-registration issues. In the current cycle 9 record, all three checks pass and the ledger caveat is resolved.

Reopen DC-005 only for one of the named semantic failures:

- Missing memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md.
- Missing TRJ-001 through TRJ-007.
- Missing C_to_B_trajectory_reuse, C_to_A_trajectory_reuse, p_survive_min, or p_verifier_reuse_min.
- Missing trajectory_node_id, replay_authorization_scope, verifier_evidence_hash, or retention_hold_state.
- Corrupt or unreadable measurement CSVs.
- Positive retained value assigned to unauthorized, stale, tampered, or retention-invalid replay.
- Sibling reuse of DC-005 TRJ-* experiment IDs or DC-005 collapse-threshold IDs.

References

No external references are cited in this cycles 7-9 clone report. REFERENCES.md exists in the workspace, but these cycles added no new literature-backed claims and did not rely on external sources beyond the project artifacts and session records.

Appendix: Implementation Details

Source Inventory

Cycle	Role	Session ID	Date	Contents
7	researcher	efcfcef0-fe2c-456f-a32a-5b2cba64dbd7	2026-05-11	Declared DC-005 closed at clone scope and directed conductor merge handling.
7	worker	5ea411ea-62ae-4821-bfc2-7c38d063fc51	2026-05-11	Built no new artifacts; listed validated DC-005 package and post-merge commands.
7	auditor	2e641351-22d1-4452-ae62-b4b066fc8f54	2026-05-11	Found and fixed the overbroad DC-006 contamination rule; validated the patch.
8	researcher	6c78a9bd-a285-4953-8efe-fae3482ecd3a	2026-05-11	Reframed the handoff around the corrected sibling-constant rule.
8	worker	d3c7f726-bb64-4d63-8094-affd9877ff3a	2026-05-11	Built no new artifacts; restated terminal handoff and corrected merge criterion.
8	auditor	e4fecfee-3f1a-4a2a-ab79-87d21226830f	2026-05-11	Revalidated DC-005 with the narrowed verifier; reported only ledger warnings.
9	researcher	8b48e188-f3bc-45d9-9c64-3649a8de253a	2026-05-11	Declared terminal merge-readiness status and no remaining clone-local work.
9	worker	82137f5f-2e2d-4c62-86a0-98d18eee9d31	2026-05-11	Built no new artifacts; confirmed conductor merge reconciliation readiness.

Cycle	Role	Session ID	Date	Contents
9	auditor	b166460d-2d8a-4a57-b87f-d79dc49e5f8c	2026-05-11	Final validation: verifier PASS, org_check PASS, promise_check PASS with 66 ledger events.

Artifact Inventory

Artifact	Lines	Purpose
memory-centric-agentive/experiments/trajectory_reuse_measurement_plan.md	298	Required DC-005 production trajectory reuse measurement plan.
tests/verify_dc005_merge_ready.py	212	Deterministic merge-readiness verifier for DC-005 parent harness rows.
memory-centric-agentive/experiments/dc005_merge_verification.md	54	Conductor-facing verifier documentation and reopen criteria.
data/dc005_merge_verification_results.csv	11	Current verifier output: 10 PASS rows plus header.
MANIFEST.md	snapshot	Workspace manifest updated to reflect the 212-line verifier and 7,376 total script lines.

Parent Harness Row Counts

The current parent measurement harness contains multiple constants. DC-005 is one branch within that shared harness.

File	Total data rows	DC-005 role
data/measurement_experiment_specs.csv	18	Contains TRJ-001 through TRJ-007.
data/measurement_required_fields.csv	22	Contains required trajectory, replay, verifier, authorization, and retention fields.
data/measurement_thresholds.csv	14	Contains the four DC-005 collapse and minimum-reuse thresholds.
data/measurement_claim_update_matrix.csv	22	Contains DC-005 claim-update rows for trajectory reuse evidence.

File	Total data rows	DC-005 role
data/measurement_synthetic_probe_results.csv	12	Contains DC-005 mechanism probes for Option C collapse behavior.

Current Verifier Results

data/dc005_merge_verification_results.csv records these 10 passing checks:

Check	Status
experiment_specs_parseable	PASS
required_fields_parseable	PASS
thresholds_parseable	PASS
claim_update_matrix_parseable	PASS
synthetic_probe_results_parseable	PASS
required_trj_ids	PASS
required_thresholds	PASS
required_fields	PASS
no_dc006_measurement_rows	PASS
authorization_gate_boundary	PASS

Cross-Reference Map

Origin	Consuming artifact	Flow
memory-centric-agentive/experiments/trajectory_reuse_measurement_plan.md	data/measurement_experiment_specs.csv	Defines TRJ-001 through TRJ-007.
memory-centric-agentive/experiments/trajectory_reuse_measurement_plan.md	data/measurement_required_fields.csv	Defines production fields for trajectory attribution, replay authorization, verifier evidence, and retention.
memory-centric-agentive/experiments/trajectory_reuse_measurement_plan.md	data/measurement_thresholds.csv	Defines C-to-B, C-to-A, branch-survival, and verifier-reuse thresholds.
data/measurement_*.csv	tests/verify_dc005_merge_ready.py	Verifier checks parseability, required rows, required thresholds, required fields, and sibling contamination boundaries.
tests/verify_dc005_merge_ready.py	data/dc005_merge_verification_results.csv	Verifier emits one PASS/FAIL row per merge-readiness check.
tests/verify_dc005_merge_ready.py	memory-centric-agentive/experiments/dc005_merge_verification.md	Documentation explains commands, checks, ledger interpretation, and reopen criteria.

Origin	Consuming artifact	Flow
Cycle 7 audit patch	tests/verify_dc005_merge_ready.py	Narrowed DC-006 rejection to reuse of DC-005 TRJ-* IDs or DC-005 collapse thresholds.
Cycle 9 audit	conductor merge decision	Confirms verifier, organization, and promise-ledger state are all green.